

**12.1.** Determine the tangent plane of the hyperboloid

$$\mathcal{H}_{2,3,1}^1 : \frac{x^2}{4} + \frac{y^2}{9} - \frac{z^2}{1} = 1$$

at the point  $M(2, 3, 1)$ . Show that the tangent plane intersects the surface in two lines.

**12.2.** Determine the generators of the hyperboloid

$$\frac{x^2}{36} + \frac{y^2}{9} - \frac{z^2}{4} = 1$$

which are parallel to the plane  $x + y + z = 0$ .

**12.3.** Determine the intersection of the hyperboloid

$$\mathcal{H}_{2,1,3}^2 : \frac{x^2}{4} + \frac{y^2}{1} - \frac{z^2}{9} = -1 \quad \text{with the line} \quad \ell : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 6 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}.$$

Write down the equations of the tangent planes at the intersection points.

**12.4.** Determine the intersection of the paraboloid

$$\mathcal{P}_{2,\frac{1}{2}}^h : x^2 - 4y^2 = 4z \quad \text{with the line} \quad \ell : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}.$$

Write down the equations of the tangent planes at the intersection points.

**12.5.** Determine the tangent planes of the elliptic paraboloid  $\frac{x^2}{5} + \frac{y^2}{3} = z$  which are parallel to the plane  $x - 3y + 2z - 1 = 0$ .

**12.6.** Determine the generators of the paraboloid

$$4x^2 - 9y^2 = 36z$$

containing the point  $P(3\sqrt{2}, 2, 1)$ .

**12.7.** Determine the generators of the paraboloid

$$\frac{x^2}{16} - \frac{y^2}{4} = z$$

which are parallel to the plane  $3x + 2y - 4z = 0$ .

**12.8.** Use a parametrization of a parabola and a rotation matrix to deduce a parametrization of an elliptic paraboloid of revolution.

**12.9.** For the surface  $\mathcal{S}$  with parametrization

$$\mathcal{S} : \begin{cases} x = \sqrt{1+t^2} \cos(s) \\ y = \sqrt{1+t^2} \sin(s) \\ z = 2t \end{cases}$$

- Give an equation of  $\mathcal{S}$ .
- Find the parameters of the point  $P(1, 1, 2)$ .
- Calculate a parametrization of the tangent plane  $T_P\mathcal{S}$  using partial derivatives.
- Give an equation of  $T_P\mathcal{S}$ .

**12.10.** For the surface  $\mathcal{S}$  with parametrization

$$\mathcal{S} : \begin{cases} x = s \\ y = t \\ z = s^2 - t^2 \end{cases}$$

- Give an equation of  $\mathcal{S}$ .
- Find the parameters of the point  $P(1, 1, 0)$ .
- Calculate a parametrization of the tangent plane  $T_P\mathcal{S}$  using partial derivatives.
- Give an equation of  $T_P\mathcal{S}$ .